Algebra SEP Worksheet 2 7/21/22

Linear Algebra

Warm Up

∗ **1.**

Let K be a field, and let A be an $n \times n$ matrix over K. Suppose that $f(x) \in K[x]$ is an irreducible polynomial such that $f(A) = 0$. Show that $\deg(f) \mid n$.

2.

Give two examples of matrices in Jordan form with minimal polynomial $x(x-1)^2(x-2)$ and characteristic polynomial $x^2(x-1)^4(x-2)$.

Work Out

3.

Let V be a finite-dimensional vector space over a field. Let $A : V \to V$ be an endomorphism, and suppose $W \subseteq V$ is an invariant subspace (so that $A(W) \subseteq W$). Let $\overline{A} : V/W \to V/W$ be defined by $\overline{A}(v + W) = Av + W$ for all $v \in V$.

(a) Show that $\overline{A}$ is well-defined.

(b) Show that if $\overline{A}$ is not invertible, then A is not invertible.

∗ **4.**

Let V be a finite-dimensional vector space over a field. Let $A : V \to V$ be a linear transformation.

(a) Show that there exists $B : V \to V$ such that $ABA = A$.

(b) Show that there exists $C : V \to V$ such that $ACA = A$ and $CAC = C$.

∗ **5.**

Let F be a field, and let V be a finite-dimensional vector space over F that has dimension $n \geq 1$. Let q be a nonzero element of F such that $q^i \neq 1$ for $1 \leq i \leq n$. Let $X : V \to V$ and $Y : V \to V$ be two F-linear transformations such that $XY = qYX$. Show that XY is nilpotent.

6.

Let V, W be two finite-dimensional vector spaces over $\mathbb{C}$. Let $A : V \to V$ and $B : V \to V$ be linear transformations. List the eigenvalues of $A \otimes B : V \otimes W \to V \otimes W$ in terms of the eigenvalues of A and B.

∗ **7.**

Let X be a subspace of $M_n(\mathbb{C})$, the $\mathbb{C}$-vector space of all $n \times n$ complex matrices. Assume that every nonzero matrix in X is invertible. Prove that $\dim_{\mathbb{C}}(X) \leq 1$.

∗ **8.**

Let $n > 0$ be an integer. Let F be a field of characteristic zero, V a vector space over F of dimension n, and $T : V \to V$ an invertible F-linear map such that $T^{-1} = T$. Denote by W the vector space of linear transformations from V to V that commute with T. Find a formula for $\dim(W)$ in terms of n and the trace of T.

∗ **9.**

Let F be a field and n be a positive integer. Fix an $n \times n$ matrix S over F this is invertible and symmetric. Writing A^t for the transpose of a matrix, we let

$$V := \{n \times n \text{ matrices } A \text{ over } F | A^t = SAS^{-1}\}.$$

Note that V is a vector space (you do not need to prove this). Find the dimension of V in terms of n.